

Gear Wall Element C

Priority	Requirement	Description	What and how can the device be measured?	Why is this important? (based on research)
1.(High)	Safe	Zero potential to cause injuries to students or teachers. Use of non-toxic, smooth, and durable materials. No sharp edges or detachable small parts.	Measure through safety certifications (e.g., ASTM, CPSIA) and user observation. 80% of child injuries from toys involve choking or sharp edges (CPSC).	Safety is the foundation of any product designed for children, especially those with special needs who may be more vulnerable to injuries. Ensuring zero risk of harm creates a trustworthy and worry-free environment for both children and caregivers.
2.(High)	Engaging	Captures and sustains children's attention for at least 15 minutes with visual, auditory, and tactile stimulation.	Conduct user engagement tests (e.g., time spent interacting) and gather feedback. Engaged children spend 30-50% more time on educational activities.	Engagement ensures children are motivated to interact with the gear wall. For special needs children, sustained engagement can support longer periods of focused learning and play, which are critical for development.
3.(High)	Conducive to Cause & effect Learning	Demonstrates clear mechanical interactions through at least 2 inputs, which lead to 4 possible outputs.	Evaluate through structured play sessions and assess children's ability to predict outcomes. 75% of cognitive development occurs in early childhood through play.	Understanding cause and effect is a fundamental cognitive skill. It helps children grasp how their actions influence their environment, promoting problem-solving, logical thinking, and curiosity.
4.(High)	Accessibility	Every life skills kid is able to interact with the device, including those with limited mobility, vision impairments, or cognitive challenges.	Use ADA compliance checks, height accessibility tests, and feedback from users. 1 in 6 children has a developmental disability affecting mobility or vision.	Special needs children have diverse abilities, and an accessible design ensures all children can benefit from the gear wall, regardless of physical, cognitive, or sensory challenges.
5.(High)	Fine motor skills	Promotes hand-eye coordination and the use of small muscles in the hands through turning, gripping, and manipulating	Measure through skill assessment tools and observation of improved dexterity. 60% of special needs	Fine motor skills are crucial for everyday tasks like writing, buttoning clothes, and using utensils. The gear wall offers a

		gears.	children benefit from fine motor skill interventions.	playful way to practice and improve these skills.
6.(Medium)	Cost effective	Designed and produced within budget constraints while maintaining quality and functionality.	Measure through cost-benefit analysis and lifecycle cost review. Schools spend 15-20% of their budgets on special needs learning tools.	Our class has been given a stringent budget to create our projects. Until we know how much other groups are spending we must keep our budget low in order to ensure all groups have adequate funding.
7.(Medium)	Customizable	Ability to rearrange or replace parts to suit different developmental levels and interests.	Test different configurations and measure user satisfaction. Customizable tools are 40% more likely to be reused in diverse settings.	Every child has unique needs and interests. A customizable design allows caregivers and educators to tailor the gear wall to individual developmental goals and preferences.
8.(Medium)	Durable	Built to withstand frequent and heavy use, including rough handling.	Evaluate through stress testing and material fatigue analysis. 70% of school equipment failures are due to lack of durability.	Special needs children may engage in repetitive or rough play, which could wear down or even break devices. Durability ensures longevity and reduces replacement costs.
9.(Low)	Unobtrusive	Does not disrupt the classroom environment or bother teachers. Is not a source of nuisance noise or light.	Measure through size constraints and user environment surveys. Space-saving designs reduce clutter by up to 25% in classrooms.	In classrooms or therapy rooms, space is often limited. An unobtrusive design ensures that the gear wall fits seamlessly into its environment without overwhelming the space or distracting from other activities.
10.(Low)	Maintainable	Easy to clean, repair, or replace components. Designed for long-term use with minimal maintenance requirements.	Assess through cleaning time trials and maintenance logs. 80% of schools prefer low-maintenance products to reduce operational costs.	Ease of maintenance ensures the product remains in good condition with minimal effort. This is especially important in high-use environments like schools and sensory rooms.

Dr. Sabatino response #1:

Hey guys -

Thank you for the info! I'm Ccing Mrs. Brennan as she is the classroom teacher and will have feedback and the final say on any structures. I'll connect with her and get back to you soon.

MT

Dr. MaryTherese Sabatino
Speech-Language Pathologist

Dr. Sabatino response #2:

Hey guys -

I saw Mr. Muscarella at a robotics things over the weekend and he asked that I reach out with some more specific info/feedback. Please see below!

We wanted to share the design requirements for the gear wall intended for the sensory room, which we have been working on. Below is a summary of the key priorities that have been identified to ensure the wall is both effective and safe for its intended users.

1. Safety: The gear wall must be designed with zero potential for injury, using non-toxic, durable materials and ensuring no sharp edges or small parts that could pose a risk.

PERFECT - our students need to be able to engage safely and there needs to be a way to ensure that adults could remove the gear/s (for cleaning, etc.) but students cannot. Some of them are STRONG :)

2. Engagement: The design should be engaging, incorporating visual, auditory, and tactile elements to capture and sustain children's attention.

Remember visual needs - high contrast colors are best. Please avoid flashing lights as we have students with seizure disorders.

3. Cause & Effect Learning: It should demonstrate clear cause-and-effect interactions, aiding children's cognitive development through their actions and the outcomes.

PERFECT

4. Accessibility: The product must be accessible to children with a wide range of abilities, including those with mobility or sensory challenges, adhering to ADA standards.

Yes - I love that you're looking into and considering ADA standards. If you have specific questions about accessibility for our specific students, please reach out. We have physical, visual, communication, mobility, and cognitive needs in the classroom.

5. Fine Motor Skills: The gear wall will facilitate the development of fine motor skills, supporting hand-eye coordination through interactive play.

Consideration of what "tasks" will be involved in access to the gears? Are the students going to be able to only move the gears, or swap them out, etc.?

6. Cost-Effectiveness: While ensuring high quality and functionality, the design must be affordable to fit within the budget constraints of educational or therapeutic environments.

consider replacement parts too - how much will it cost to repair/replace parts if needed?

7. Customizability: The wall should be customizable to accommodate different developmental levels and preferences, allowing for flexibility in use.

I think consideration of ways to turn off and on different sensory parts (e.g., if a kid enjoys the movement but cannot handle sounds, etc.)

8. Durability: The design must be durable enough to withstand frequent use and rough handling, ensuring longevity and minimal maintenance. YES. YEAH. YUUP.

9. Unobtrusive: It should be compact and unobtrusive, fitting seamlessly into its environment without overwhelming other tools or activities.

consider something that can easily block the toy from view if needed? Like a drop-down curtain or something that can visually mask if needed?

10. Maintainability: The product should be easy to clean and maintain, ensuring its longevity with minimal effort.
100%.

THANK YOU!!!!

Dr. MaryTherese Sabatino
Speech-Language Pathologist